A Constructive Introduction to Quantum Field Theory

Lecture 4: The Polchinski flow and forward–backward equations

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Abstract

Building on the scale decomposition and the stochastic observables of Lecture 3, we introduce interactions through the Polchinski equation for the effective potential. Its probabilistic counterpart is a forward–backward stochastic differential equation (FBSDE) for the field seen at increasing resolution, whose effective force is a martingale. Writing the FBSDE in the coordinate given by the conditional mean of the terminal field removes its drift and integrates the transport part of the force flow exactly. In the gradient case the remaining equation for the force is the Wetterich equation for the effective average action, whose propagator is the conditional covariance. We then discuss the variational structure of the flow (Boué–Dupuis formula and Hamilton–Jacobi–Bellman equation) in finite and infinite volume. The additional material covers the algebraic and perturbative structure of the flow (semigroup, flatness, counterterms for Φ^4_3), and the Dyson–Schwinger equations at every scale. The treatment is formal: we compute as if all objects were smooth and integrable, and we indicate where the real difficulties lie.

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Introduction

In Lecture 3 we decomposed the Gaussian free field along a resolution parameter and took as primary objects the stochastic observables: martingales in the filtration of finite-resolution measurements [BCG26]. In this lecture we introduce interactions. The effective potentials solve the Polchinski equation, and the interacting field seen at increasing resolution solves a forward–backward stochastic differential equation whose force is itself an observable. See also [Pol84, BK87, Sal99, BBD24] for the renormalisation group side and [BG20, BG24] for the variational side.

Structure. Part I (Sections 7–9) is the core of the lecture: the Polchinski equation and its FBSDE, the FBSDE in force coordinates and the Wetterich equation, and the variational structure. Part II (Sections 10 and 11) is additional material, referred to from Part I where needed.

What is formal. Rigorous statements require a finite-dimensional setting: a finite lattice, or a Fourier cutoff on the torus, together with a finite terminal resolution T. There, all the identities below are theorems under mild growth assumptions on the interaction, and their proofs are the computations we give. The hard problems are the limits $T \to \infty$ (ultraviolet) and infinite volume (infrared), together with the locality of the limiting objects. We mark with (formal) the statements that we only derive at this level.

Conventions. As in Lectures 1–3: $\mathcal{M} = \mathbb{T}_L^d$ or $\mathbb{R}^d$, $A = m^2 - \Delta$, $C = A^{-1}$ with kernel G, μ_m the GFF, $\langle f, g\rangle = \int fg$ for real functions. For a functional F on smooth fields, $DF(\varphi) \in \mathcal{S}'(\mathcal{M})$ and $D^2F(\varphi)$ are its first and second derivatives, $D_hF = \langle DF, h\rangle$. $H_n(a;c)$ are the Hermite polynomials of variance c, and μ^Q is the centred Gaussian measure with covariance Q.

Part I. Main lecture

7 Interactions: the Polchinski equation and FBSDEs

7.1 Effective potentials

Fix a finite volume $\mathcal{M} = \mathbb{T}_L^d$ and a terminal resolution $T < \infty$, which plays the role of an ultraviolet cutoff: the reference field $X_T \sim \mu^{Q_T}$ is smooth. Let V_T be an interaction, a functional of

smooth fields bounded below. Our main example is

$$V_T(\varphi) = \int_{\mathbb{T}_L^d} [\lambda H_4(\varphi(x); c_T) + \gamma_T H_2(\varphi(x); c_T)]\mathrm{d}x, \tag{7.1}$$

with a mass counterterm γ_T to be fixed in Section 10.2 ($\gamma_T = 0$ for $d = 2$). The interacting measure at cutoff T is

$$\nu_T(\mathrm{d}\varphi) = Z_T^{-1} \mathrm{e}^{-V_T(\varphi)} \mu^{Q_T}(\mathrm{d}\varphi). \tag{7.2}$$

Definition 7.1. *The* effective potential *at resolution $t \in [0,T]$ is*

$$\mathrm{e}^{-V_t(\varphi)} := \int \mathrm{e}^{-V_T(\varphi+\zeta)}\, \mu^{Q_{T,t}}(\mathrm{d}\zeta). \tag{7.3}$$

V_t is the interaction felt by the field at resolution t once the fluctuations between t and T have been integrated out: it is the Wilson effective action. At $t = 0$, $V_0(0) = -\log Z_T$ is the free energy. Here is the first key fact.

Proposition 7.2. (Marginals) *Let $\mathbb{P}$ be defined by $\mathrm{d}\mathbb{P}/\mathrm{d}\mathbb{P}_0 = Z_T^{-1}\mathrm{e}^{-V_T(X_T)}$, where $\mathbb{P}_0$ is the law of the Gaussian process X. Then $\mathrm{Law}_{\mathbb{P}}(X_T) = \nu_T$, and for every $t \le T$*

$$\mathrm{Law}_{\mathbb{P}}(X_t) = \nu_t, \qquad \nu_t(\mathrm{d}\varphi) := Z_T^{-1} \mathrm{e}^{-V_t(\varphi)} \mu^{Q_t}(\mathrm{d}\varphi). \tag{7.4}$$

Moreover, the conditional law of X_T given $\mathcal{F}_t$ is

$$\mu_t^{X_t}(\mathrm{d}\varphi), \qquad \mu_t^{\phi}(G) := \mathrm{e}^{V_t(\phi)} \int G(\phi+\zeta)\, \mathrm{e}^{-V_T(\phi+\zeta)}\, \mu^{Q_{T,t}}(\mathrm{d}\zeta). \tag{7.5}$$

Proof. Under $\mathbb{P}_0$, $X_T = X_t + (X_T - X_t)$ with an independent increment of law $\mu^{Q_{T,t}}$. Hence $\mathbb{E}_{\mathbb{P}}[G(X_t)] = Z_T^{-1}\, \mathbb{E}_0[G(X_t)\mathrm{e}^{-V_T(X_T)}] = Z_T^{-1}\, \mathbb{E}_0[G(X_t)\mathrm{e}^{-V_t(X_t)}]$. The conditional law follows in the same way. □

So at every resolution the field is again a Gibbs measure, with reference covariance Q_t and potential V_t. The measure-valued martingale $(\mu_t^{X_t})_t$ goes from ν_T at $t = 0$ to δ_{X_T} at $t = T$: it is a *stochastic localisation* of ν_T in the sense of Eldan [Eld23], adapted to the covariance profile [BBD24, BCG26].

Theorem 7.3. (Polchinski equation [Pol84]) (formal) *V_t solves the backward Hamilton–Jacobi equation*

$$\partial_t V_t = \tfrac{1}{2}\langle D V_t, \dot{Q}_t\, D V_t\rangle - \tfrac{1}{2}\mathrm{Tr}(\dot{Q}_t\, D^2 V_t), \qquad t \in [0,T], \tag{7.6}$$

with terminal condition V_T.

Proof. $u_t := \mathrm{e}^{-V_t} = \mathrm{e}^{\frac{1}{2}\Delta_{Q_{T,t}}} u_T$ is a Gaussian convolution, where $\Delta_Q := \mathrm{Tr}(Q\, D^2)$. Since $\partial_t Q_{T,t} = -\dot{Q}_t$, it solves the backward heat equation $\partial_t u_t = -\tfrac{1}{2}\mathrm{Tr}(\dot{Q}_t D^2 u_t)$. Insert $D^2 u = (D V \otimes D V - D^2 V)\, u$ and divide by $-u$. □

The nonlinear term generates new, generally nonlocal, interactions (trees), and the trace term contracts pairs of fields with the covariance of the shell $[t, t+\mathrm{d}t]$ (loops). The substitution $u = \mathrm{e}^{-V}$, which linearises the equation, is a Cole–Hopf transform: we come back to this in Section 7.4, and in Section 10.

7.2 The Polchinski process

Theorem 7.4. (FBSDE for the interacting field) (formal) *Under $\mathbb{P}$ the process $\Phi_t := X_t$, the* interacting field at resolution t, *solves*

$$\mathrm{d}\Phi_t = \dot{Q}_t\, F_t\, \mathrm{d}t + \dot{Q}_t^{1/2}\, \mathrm{d}W_t^{\mathbb{P}}, \qquad F_t = -D V_t(\Phi_t) = \mathbb{E}_t^{\mathbb{P}}\,[-D V_T(\Phi_T)], \tag{7.7}$$

with $\Phi_0 = 0$ and a $\mathbb{P}$-cylindrical Brownian motion $W^{\mathbb{P}}$. In particular the effective force $(F_t)_{t \le T}$ *is a $\mathbb{P}$-martingale: an observable field of the interacting theory.*

Proof. The density process is $M_t := \mathbb{E}_0[\mathrm{d}\mathbb{P}/\mathrm{d}\mathbb{P}_0|\mathcal{F}_t] = Z_T^{-1} \mathrm{e}^{-V_t(X_t)}$, by Proposition 7.2. By Itô's formula and (7.6), its drift vanishes and $\mathrm{d}M_t = -M_t \langle D V_t(X_t), \dot{Q}_t^{1/2} \mathrm{d}W_t \rangle$. By Girsanov's theorem, $W_t^{\mathbb{P}} = W_t + \int_0^t \dot{Q}_s^{1/2} D V_s(X_s) \mathrm{d}s$ is a $\mathbb{P}$-Brownian motion, which is (7.7). Differentiating (7.3) gives $D V_t(\phi) = \mu_t^{\phi}(D V_T)$, which is the second identity. Alternatively, Itô's formula for $D V_t(\Phi_t)$, together with the derivative of (7.6), shows that its drift vanishes. Here one uses that $D^2 V_t$ is symmetric. □

Read as an equation for an unknown pair (Φ, F), (7.7) is a *forward–backward* system [MY99]. The field is driven forward in resolution by a force that is determined backward, as the conditional expectation of a functional of the terminal field:

$$\Phi_t = \int_0^t \dot{Q}_s F_s \, \mathrm{d}s + \int_0^t \dot{Q}_s^{1/2} \mathrm{d}W_s, \qquad F_t = \mathbb{E}_t[-D V_T(\Phi_T)], \qquad t \in [0, T]. \tag{7.8}$$

Here and below we drop the superscript and write W for the Brownian motion driving the interacting field. The Polchinski equation is not needed to state (7.8), and $\nu_T = \mathrm{Law}(\Phi_T)$ is recovered from the solution. This is the starting point of FBSDE constructions of Euclidean fields (Figure 7.1): one solves (7.8) with renormalised V_T, with estimates uniform in T and in the volume, and takes the limit. The field coordinate Φ is *not* an observable, only a semimartingale. Its martingale part is the Gaussian noise, and the observables are built from it and from F.

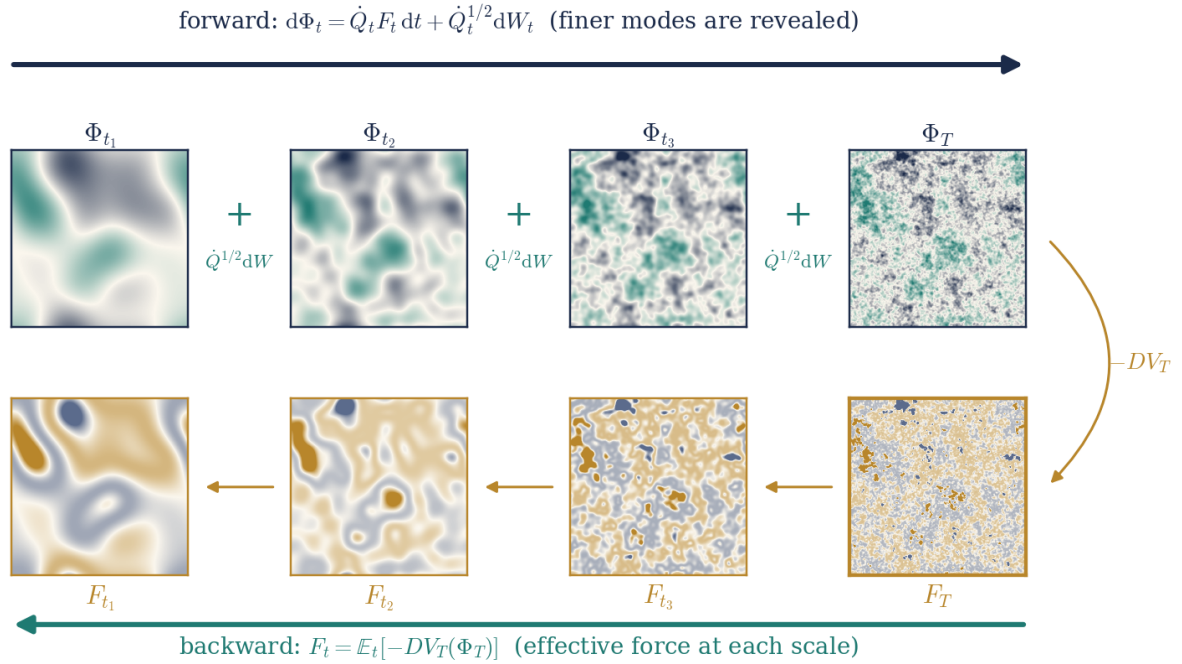


Figure 7.1. The forward–backward structure of (7.8) on $\mathbb{T}^2$, heat profile, at resolutions $t_1 < t_2 < t_3 < T$, all built from the same noise. Top: the field gains the fluctuation of each shell, $\dot{Q}_t^{1/2} \mathrm{d}W_t$, and reveals finer modes. Bottom: the effective force $F_t = \mathbb{E}_t[-D V_T(\Phi_T)]$ for the Φ^4 interaction, determined backward from the terminal interaction. It is shown to first order in λ, where it is the Wick cube $-4\lambda H_3(\Phi_t; c_t)$, and each panel is in its own units.

Remark 7.5. **(Closed versus local)** At finite T the force is a closed observable, $F_t = \mathbb{E}_t[F_T]$. For Φ^4, $F_T(x) = -4\lambda H_3(\Phi_T(x); c_T) - 2\gamma_T H_1(\Phi_T(x); c_T)$ is local. As $T \to \infty$ one expects the force to converge as an observable. In $d = 2$ the limit is closed, $F_\infty = -4\lambda{:}\Phi_\infty^3{:}$. In $d = 3$ it is expected not to be. If it were, $F_t = \mathbb{E}_t[F_\infty]$ with $F_\infty \in L^1$, the DS equation of Section 11 would pass to the limit ($T \to \infty$, then $t \to \infty$), with F_∞ paired only with functionals of Φ_∞. Conditioning on $\sigma(\Phi_\infty)$ would then give a logarithmic derivative of the Φ_3^4 measure that is an integrable function of the field, $\langle A h, \Phi_\infty\rangle - \mathbb{E}[F_\infty(h)|\Phi_\infty]$. This is not expected: the regularised logarithmic derivatives do not converge in L^2 [ALZ06]. The limiting force would then be an observable in the proper sense, a compatible family with no terminal representative. Locality of the limiting force is the crux of the construction [BCG26].

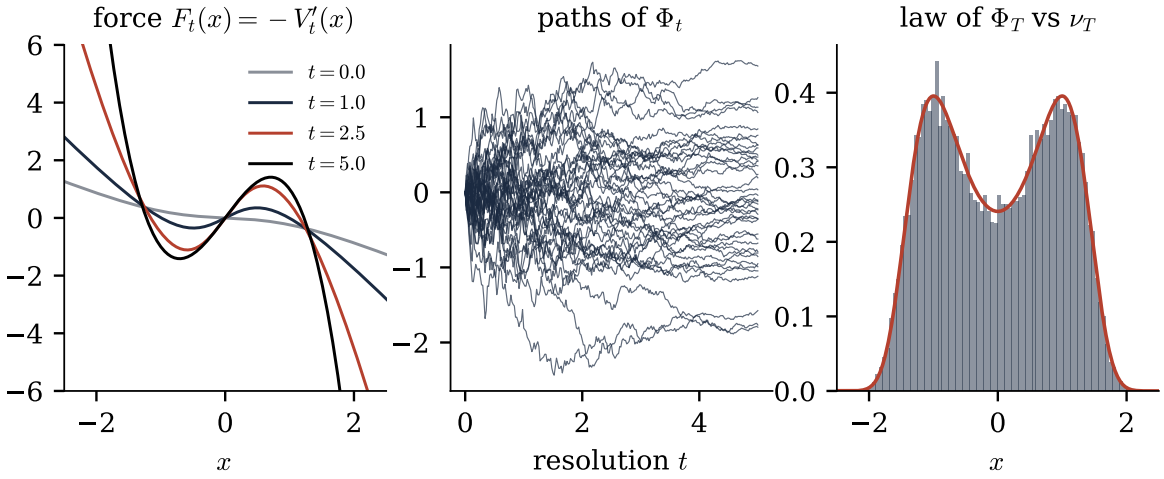


Figure 7.2. A zero-dimensional Polchinski process (Exercise 7.2): $Q_t = 1 - \mathrm{e}^{-t}$, $T = 5$, $V_T(x) = \frac{1}{2}x^4 - \frac{3}{2}x^2$. Left: the effective force $F_t = -V_t'$ is almost linear at low resolution and develops the double-well structure as $t \to T$. Centre: paths of the FBSDE (7.7), which split between the two wells (stochastic localisation). Right: histogram of Φ_T ($2 \cdot 10^4$ paths) against the density of ν_T.

7.3 Observables of the interacting theory

For a functional G of the terminal field, the observable $O_t = \mathbb{E}_t^{\mathbb{P}}[G(\Phi_T)]$ is $O_t = G_t(\Phi_t)$, with

$$G_t(\phi) = \mu_t^{\phi}(G) = \mathrm{e}^{V_t(\phi)} \int G(\phi + \zeta)\, \mathrm{e}^{-V_T(\phi+\zeta)} \mu^{Q_{T,t}}(\mathrm{d}\zeta). \tag{7.9}$$

Formally, G_t solves the backward Kolmogorov equation of the interacting dynamics,

$$\partial_t G_t + \tfrac{1}{2}\mathrm{Tr}(\dot{Q}_t D^2 G_t) - \langle \dot{Q}_t D V_t, D G_t\rangle = 0, \qquad G_T = G, \tag{7.10}$$

which is the Doob transform of the Gaussian heat equation by e^{-V_t} (Exercise 7.3). Conversely, any solution of (7.10) yields an observable $G_t(\Phi_t)$; at finite T it is closed, but as $T \to \infty$ it need not be. The force is the case $G = -D V_T$, and is special because it also drives the dynamics.

Tangent observables. Let V_T^θ depend on a coupling θ, and let $U_t := \partial_\theta V_t^\theta$. Differentiating (7.3) gives $U_t(\Phi_t) = \mathbb{E}_t^\theta[\partial_\theta V_T^\theta(\Phi_T)]$, an observable, and

$$\partial_\theta \mathbb{E}^{\nu_T^\theta}[G] = -\mathrm{Cov}^{\nu_T^\theta}(G, \partial_\theta V_T^\theta). \tag{7.11}$$

Thus the tangent observables, such as the running $:\varphi^4:$ for $\theta=\lambda$ (together with the derivative of the counterterm $\partial_\lambda \gamma_T \int :\varphi^2:$), describe the response of the theory to a change of coupling. They are the observables conjugate to the coordinates of theory space.

7.4 The force and a Burgers equation

Differentiating (7.6), the effective force $F_t=-D V_t$, now as a function of the field, solves

$$\partial_t F_t = -D F_t[\dot{Q}_t F_t] - \tfrac{1}{2}\mathrm{Tr}(\dot{Q}_t D^2 F_t), \qquad F_T = -D V_T, \tag{7.12}$$

an infinite-dimensional, backward, viscous *Burgers equation*. The Cole–Hopf transform $F = D\log u$ linearises it, which is the statement that $u=\mathrm{e}^{-V}$ solves a heat equation. Two remarks on (7.12):

i. it uses only the force, which is a local quantity (*intensive*), while V_t is *extensive*, of order of the volume. This matters in infinite volume (Section 9.2);

ii. the Burgers flows for different covariances commute, and the curvature vanishes, when $D F_t$ is symmetric, i.e. when F_t is a gradient [BCG26]. For a non-gradient force, as for the general Wilson–Itô diffusions of [BCG26], the flows do not commute.

Exercises

Exercise 7.1. (Quadratic interactions) Let $V_T(\varphi)=\frac{\kappa}{2}\langle\varphi,\varphi\rangle$, $\kappa>0$. (a) Show that $V_t(\varphi)=\frac{1}{2}\langle\varphi,\kappa(1+\kappa Q_{T,t})^{-1}\varphi\rangle+$ const solves (7.6). (b) Show that ν_T is Gaussian with covariance $(Q_T^{-1}+\kappa)^{-1}$: for $Q_T\to(m^2-\Delta)^{-1}$ this is the GFF of mass $(m^2+\kappa)^{1/2}$. (c) Write (7.7) explicitly. It is a linear FBSDE. Solve it, and check that F_t is a martingale. *Moral:* the mass can be put either into the covariance or into the force.

Exercise 7.2. (Zero dimensions) Take $\mathcal{M}$ a point, $Q_t=q(t)$ scalar, increasing, $V_T(x)=\lambda x^4-a x^2$ with $\lambda>0$, $a\geq 0$ (a double well if $a>0$; Figure 7.2 has $\lambda=\frac{1}{2}$, $a=\frac{3}{2}$). (a) Write (7.6) and solve it numerically for $q(t)=1-\mathrm{e}^{-t}$. (b) Simulate (7.7) by first computing V_t' on a grid, and compare the histogram of Φ_T with $\mathrm{e}^{-\lambda x^4+a x^2-x^2/2q(T)}$. (c) Compute V_t to second order in λ.

Exercise 7.3. (Doob transform) Show that G_t in (7.9) solves (7.10), by writing $\mathrm{e}^{-V_t}G_t=\mathrm{e}^{\frac{1}{2}\Delta_{Q_{T,t}}}(\mathrm{e}^{-V_T}G)$. Prove (7.11).

8 The FBSDE in force coordinates: the Wetterich equation

In this section the forward–backward system (7.8) is the primary object. We look for the coordinates in which it is simplest. They remove the predictable part of the forward dynamics, and they integrate exactly the bilinear (tree) part of the force flow (7.12). In the gradient case the resulting equation for the force is the Wetterich equation for the effective average action [Wet93, BTW02]. Throughout,

$$S_t := Q_T - Q_t, \qquad \dot{S}_t = -\dot{Q}_t, \qquad S_T = 0,$$

is the covariance still unresolved at resolution t, and (Φ, F) solves (7.8) with a martingale force $F_t=F_t(\Phi_t)$, $\mathrm{d}F_t=\Upsilon_t\,\mathrm{d}W_t$.

8.1 State and costate

Pontryagin's maximum principle. Consider first a deterministic problem: minimise the cost $J(u) = g(x_T) + \int_t^T \ell(x_s, u_s)\,\mathrm{d}s$ over controls u on $[t,T]$, where the state solves $\dot{x}_s = b(x_s, u_s)$, $x_t = x$. With the *Hamiltonian*

$$H(x,p,u) := \langle p, b(x,u)\rangle - \ell(x,u),$$

Pontryagin's principle [PBGM62] says that along an optimal pair (x,u) there is a *costate* p such that

$$\dot{x}_s = \partial_p H, \qquad \dot{p}_s = -\partial_x H, \qquad p_T = -Dg(x_T), \qquad u_s \in \arg\max_v H(x_s, p_s, v).$$

The state runs forward from its initial value, the costate backward from its terminal value: a two-point boundary value problem. In the smooth case the principle is the first-order condition. If $\xi = \partial_\varepsilon x$ is the variation of the state under $u \to u + \varepsilon v$, then $\dot{\xi} = \partial_x b\,\xi + \partial_u b\,v$, $\xi_t = 0$, and the costate equation is exactly what makes $\frac{\mathrm{d}}{\mathrm{d}s}\langle p, \xi\rangle = \langle \partial_x \ell, \xi\rangle + \langle p, \partial_u b\, v\rangle$. Integrating over $[t, T]$ and using the terminal condition,

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon} J(u + \varepsilon v)|_{\varepsilon=0} = -\int_t^T \langle \partial_u H(x_s, p_s, u_s), v_s\rangle\,\mathrm{d}s.$$

So the costate is the Lagrange multiplier of the constraint $\dot{x} = b$, and an optimal control is a critical point of the Hamiltonian.

The costate is a gradient. Let $V(t,x) = \min_u J(u)$ be the value function. Comparing the cost from (t,x) with the cost from $(t + \mathrm{d}t, x + b\,\mathrm{d}t)$ gives the Hamilton–Jacobi–Bellman equation

$$\partial_t V = H_*(x, -DV), \qquad H_*(x,p) := \max_u H(x,p,u), \qquad V(T,\cdot) = g.$$

Its characteristics are Hamilton's equations for H_*, and along them $p_s = -DV(s, x_s)$: the costate is the sensitivity of the optimal cost to a displacement of the state, i.e. the force that the future exerts on the present.

The Hamiltonian of the field. In the problem of Theorem 9.1 the state is the field, $b(x,u) = \dot{Q}_s^{1/2} u$, $\ell = \frac{1}{2}\|u\|^2$ and $g = V_T$. Then $H(p,u) = \langle p, \dot{Q}_s^{1/2} u\rangle - \frac{1}{2}\|u\|^2$ is maximal at $u = \dot{Q}_s^{1/2} p$, and

$$H_*(p) = \frac{1}{2}\langle p, \dot{Q}_s p\rangle$$

is the Hamiltonian of a free particle with mass matrix $\dot{Q}_s^{-1}$: it does not depend on the state. Hence $\dot{p} = 0$ and $\dot{x} = \dot{Q}_s p$: the costate is conserved and the state moves along the shear (8.1). All the interaction sits in the terminal condition $p = -DV_T(x + S_t p)$, an implicit equation for the momentum, solved by a sum of tree graphs (Exercise 8.2). The corresponding Hamilton–Jacobi equation $\partial_t V = \frac{1}{2}\langle DV, \dot{Q}_t DV\rangle$ is the Polchinski equation (7.6) without its trace term.

With noise. For a controlled diffusion $\mathrm{d}X = b(X,u)\,\mathrm{d}s + \sigma\,\mathrm{d}W$, with σ independent of the control, the stochastic maximum principle [Bis73, Pen90, YZ99] keeps the Hamiltonian and the maximisation, but the costate becomes an adapted process which solves a *backward* stochastic differential equation,

$$\mathrm{d}p_s = -\partial_x H\,\mathrm{d}s + Y_s\,\mathrm{d}W_s, \qquad p_T = -Dg(X_T).$$

The martingale term $Y\,dW$ is the price of adaptedness: the costate may not anticipate the noise. For the field $\partial_x H = 0$, so the costate is a martingale, $p_t = \mathbb{E}_t[-D\,V_T(X_T)]$, and since the value function is the effective potential (Theorem 9.1), $p_t = -D\,V_t(X_t)$. The noise adds the Itô term to the Hamilton–Jacobi equation, which becomes the Polchinski equation (7.6): its quadratic, Hamilton–Jacobi part generates the trees, and the trace term generates the loops. The forward–backward system (7.8) is the stochastic Hamiltonian system of this problem.

The FBSDE is the stochastic Pontryagin system of the control problem of Theorem 9.1. The controlled field $dX = \dot{Q}_t^{1/2}(u\,dt + dW)$ has Hamiltonian $\langle p, \dot{Q}_t^{1/2}u\rangle - \frac{1}{2}\|u\|^2$, which does not depend on the state. By the stochastic maximum principle (Exercise 8.1), the adjoint process p has no drift, $dp_t = Y_t\,dW_t$, with terminal value $p_T = -D\,V_T(X_T)$, and the optimal control is $u_t = \dot{Q}_t^{1/2}p_t$. Thus

the field Φ is the state, and the force F is the costate.

The function $F_t(\cdot)$ with $F_t = F_t(\Phi_t)$ is the *decoupling field* of the FBSDE, and the Burgers equation (7.12) is its quasilinear PDE; $Y_t = D\,F_t(\Phi_t)\,\dot{Q}_t^{1/2}$ is the martingale representation integrand of the costate. The two terms of (7.12) have distinct origins. The transport term $-D\,F\,[\dot{Q}\,F]$ comes from the drift of the state, and generates the tree diagrams. The diffusion term $-\frac{1}{2}\mathrm{Tr}(\dot{Q}\,D^2F)$ is the Itô correction, and generates the loops.

Without noise the state–costate pair follows the Hamiltonian flow of the free Hamiltonian $h_s(p) = \frac{1}{2}\langle p, \dot{Q}_s p\rangle$: $\dot{p} = 0$ and $\dot{x} = \dot{Q}_s p$. From t to T this flow is the linear shear

$$\Sigma_t\colon (x,p) \longmapsto (x + S_t p,\ p). \tag{8.1}$$

Its trajectories are the characteristics of the transport part of (7.12) (Exercise 8.2).

8.2 Removing the predictable drift

The drift $\dot{Q}_t F_t$ of the state is predictable, and the costate is a martingale. Hence the drift that remains to be accumulated on $[t,T]$ is predicted, at resolution t, by transporting the current state with the current costate along the free flow (8.1).

Proposition 8.1. (The forecast) (formal) *The conditional mean of the terminal field is*

$$\psi_t := \mathbb{E}_t[\Phi_T] = \Phi_t + S_t F_t, \tag{8.2}$$

i.e. $(\psi_t, F_t) = \Sigma_t(\Phi_t, F_t)$. *It is a martingale, with*

$$d\psi_t = \dot{Q}_t^{1/2} dW_t + S_t Y_t\,dW_t = (1 + S_t\,D\,F_t(\Phi_t))\,\dot{Q}_t^{1/2} dW_t. \tag{8.3}$$

Proof. Integrate the forward equation from t to T and condition on $\mathcal{F}_t$: since F is a martingale, $\mathbb{E}_t \int_t^T \dot{Q}_s F_s ds = S_t F_t$. For the differential, the drift $\dot{Q}_t F_t$ of Φ cancels against $\dot{S}_t F_t$, and $S_t\,dF_t = S_t Y_t\,dW_t$. □

So ψ is a closed observable, the *field observable* of [BCG26]. The field splits as

$$\Phi_t = \psi_t - S_t F_t\colon \tag{8.4}$$

a martingale, minus the drift still expected on the unresolved scales, which vanishes at $t=T$. The noise of ψ in (8.3) has two sources: the fresh fluctuation of the shell, $\dot{Q}_t^{1/2}\mathrm{d}W_t$, and the revision of the forecast of the remaining drift, $S_t\,\mathrm{d}F_t$. The coupling between the forward and the backward equations, which was in the drift of Φ, now sits in the volatility of ψ. The coordinate ψ is the *interaction picture* with respect to the noiseless (classical) part of the dynamics.

8.3 The FBSDE in the forecast coordinate

Assume that the map $x\mapsto\psi=x+S_tF_t(x)$ is invertible, and write its inverse as

$$x=\psi+S_tF_t(\psi),\qquad F_t(\psi):=-F_t(x),\tag{8.5}$$

so that $F_t=-F_t(\psi_t)$: the costate is now a function of the forecast. Introduce the *response* and the *regulator*

$$H_t(\psi):=(S_t^{-1}+DF_t(\psi))^{-1},\qquad R_t:=S_t^{-1}-Q_T^{-1},\qquad \dot{R}_t=S_t^{-1}\dot{Q}_tS_t^{-1}\geq 0.\tag{8.6}$$

Differentiating the two maps $x\leftrightarrow\psi$ gives $1+S_tDF_t(x)=(1+S_tDF_t(\psi))^{-1}=H_t(\psi)\,S_t^{-1}$.

Theorem 8.2. (FBSDE in force coordinates) (formal) *In the coordinates (ψ,F), the system (7.8) becomes*

$$\mathrm{d}\psi_t=H_t(\psi_t)\;S_t^{-1}\dot{Q}_t^{1/2}\,\mathrm{d}W_t,\qquad F_t(\psi_t)=\mathbb{E}_t[DV_T(\psi_T)],\tag{8.7}$$

with $\psi_0=S_0F_0=\mathbb{E}[\Phi_T]$ and $\psi_T=\Phi_T$. The forward equation has no drift, and its covariance is $C_t:=H_t\dot{R}_tH_t^*$. The decoupling field F solves the pure diffusion equation

$$\partial_tF_t+\tfrac{1}{2}\mathrm{Tr}(C_t\,D^2F_t)=0,\qquad F_T=DV_T.\tag{8.8}$$

Proof. The forward equation is (8.3) with $1+S_tDF_t=H_tS_t^{-1}$, and its covariance is

$$H_tS_t^{-1}\dot{Q}_tS_t^{-1}H_t^*=H_t\dot{R}_tH_t^*.$$

The backward equation is $F_t=\mathbb{E}_t[F_T]$ with $\psi_T=\Phi_T$. Since $F_t(\psi_t)$ is a martingale and ψ has no drift, the drift in Itô's formula for $F_t(\psi_t)$ is the left-hand side of (8.8). □

Compare (8.8) with the Burgers equation (7.12): the transport term has been absorbed exactly by the change of coordinates, and only the Itô term is left. If it is dropped, $F_t=DV_T$ does not move: all the tree diagrams are contained in the coordinate change (8.5). The loops are generated only by the quadratic variation C_t of the forecast, i.e. by the randomness of the noise and of the force revisions, and the nonlinearity has moved into the field dependence of C_t. None of this uses a gradient structure or a Gaussian convolution: Theorem 8.2 holds for any FBSDE of the form (7.8) with a Markov martingale force, gradient or not, for which the coordinate change is invertible. For a general terminal force $F_T=\mathfrak{f}(\Phi_T)$, replace DV_T by $-\mathfrak{f}$ in (8.7) and (8.8).

8.4 Conditional moments

For the Polchinski process the response has a direct meaning.

Proposition 8.3. (formal) *If $F_t=-DV_t$ then*

$$H_t(\psi_t)=\mathrm{Cov}_t:=\mathrm{Cov}^{\mathbb{P}}(\Phi_T\,|\,\mathcal{F}_t)=S_t-S_tD^2V_t(\Phi_t)\,S_t.$$

In particular $H_t > 0$ is symmetric, $1 + S_t D F_t = \mathrm{Cov}_t S_t^{-1}$ is always invertible, and $H_t \le S_t$ when V_T is convex.

Proof. By (7.5), the conditional law μ_t^{ϕ} is the tilt of $\mu^{S_t}(\cdot - \phi)$ by e^{-V_T}; differentiating twice in ϕ gives $\mathrm{Cov}(\mu_t^{\phi}) = S_t - S_t D^2 V_t(\phi)\, S_t$. On the other hand $H_t S_t^{-1} = 1 + S_t D F_t = 1 - S_t D^2 V_t$. The last claim is the Brascamp–Lieb inequality. □

So the forecast coordinates are the first two conditional moments of the stochastic localisation $\mu_t^{\Phi_t}$ of Proposition 7.2: $(\psi_t, \mathrm{Cov}_t) = (\mathbb{E}_t \Phi_T, \mathrm{Cov}_t \Phi_T)$ (Figure 8.1). They are tied by Itô's formula: since $\mathbb{E}_t[\Phi_T \otimes \Phi_T] = \psi_t \otimes \psi_t + \mathrm{Cov}_t$ is a martingale,

$$\mathrm{d}\langle \psi, \psi \rangle_t = C_t \,\mathrm{d}t = \mathrm{Cov}_t \dot{R}_t \mathrm{Cov}_t \,\mathrm{d}t, \qquad \mathrm{dCov}_t = -\mathrm{Cov}_t \dot{R}_t \mathrm{Cov}_t \,\mathrm{d}t + \mathrm{d}(\text{martingale}). \tag{8.9}$$

Conditional variance is resolved at the rate $\mathrm{Cov}_t \dot{R}_t \mathrm{Cov}_t$, which for the Gaussian reference is $S_t \dot{R}_t S_t = \dot{Q}_t$.

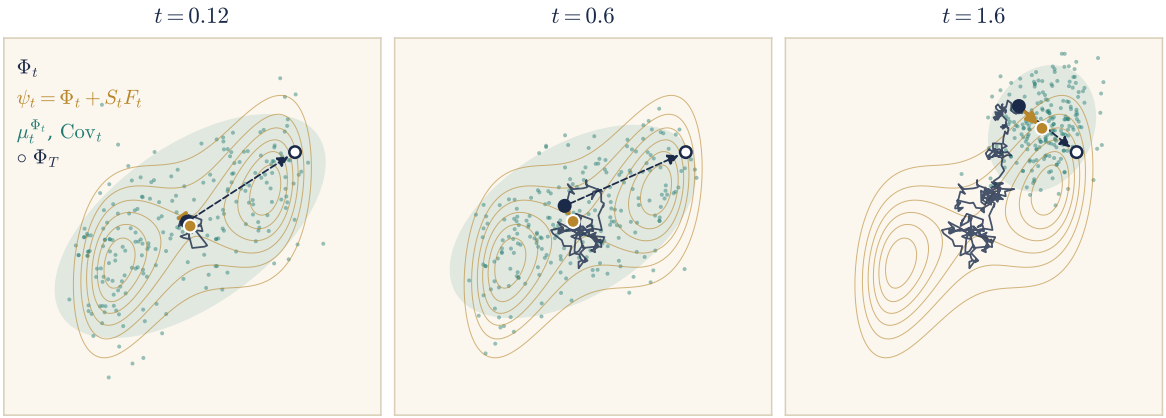


Figure 8.1. Stochastic localisation for a two-component field: $Q_t = (1 - \mathrm{e}^{-t})\,1$, $T = 4$, V_T a tilted double well, and the contours show the density of ν_T. One path of (7.7) up to resolution t (navy), with its terminal value Φ_T (open circle). The forecast $\psi_t = \Phi_t + S_t F_t$ (gold) is the mean of the conditional law $\mu_t^{\Phi_t}$: teal samples, and the ellipse is two standard deviations of Cov_t. At low resolution the conditional law covers both wells; as t grows it localises on one of them.

8.5 The gradient sector: Wetterich's equation

Now let $F_t = -D V_t$. The graph $\{(x, -D V_t(x))\}$ is a Lagrangian submanifold of the (field, force) phase space, and the shear (8.1) is symplectic because S_t is symmetric. Its image is again Lagrangian. It is a graph over ψ: the map $x \mapsto x + S_t F_t(x)$ is S_t times the gradient of $\frac{1}{2}\langle x, S_t^{-1} x \rangle - V_t(x)$, which is strictly convex since its Hessian is $S_t^{-1} \mathrm{Cov}_t S_t^{-1} > 0$ (Proposition 8.3), so the map is injective. Hence the image is the graph of a gradient:

$$F_t = D\Gamma_t, \qquad H_t = (S_t^{-1} + D^2 \Gamma_t)^{-1}, \qquad \Gamma_T = V_T. \tag{8.10}$$

The generating function of this canonical transformation is the Moreau (infimal convolution) transform

$$V_t(x) = \inf_{\psi} \Big\{ \Gamma_t(\psi) + \tfrac{1}{2} \langle x - \psi, S_t^{-1}(x - \psi) \rangle \Big\}, \tag{8.11}$$

whose optimality conditions $x = \psi + S_t D\Gamma_t(\psi)$, $D V_t(x) = D\Gamma_t(\psi)$ are (8.5). It is well defined by the same convexity: up to the change of variable $j = S_t^{-1} x$, it is a Legendre transform. We use (8.11) to fix the additive constant in Γ_t. Figure 8.2 shows the construction in zero dimensions.

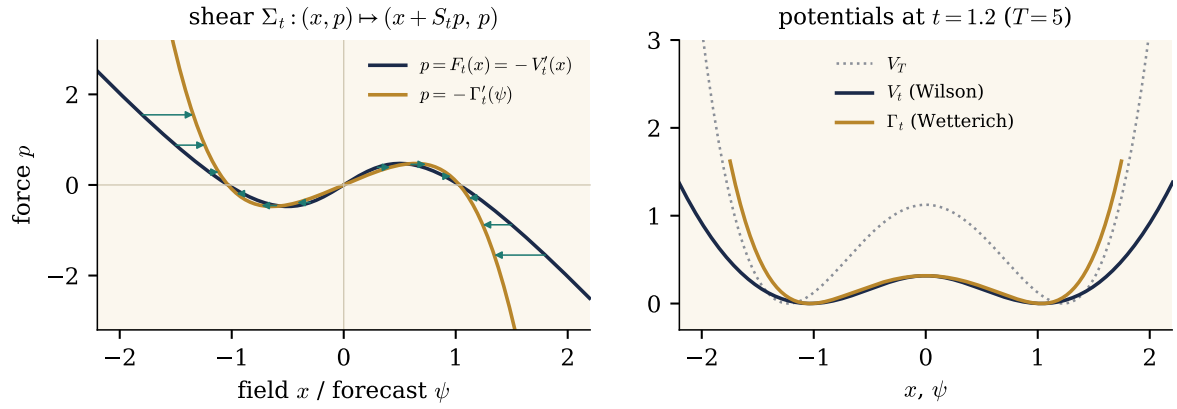


Figure 8.2. The zero-dimensional model of Figure 7.2 at $t = 1.2$. Left: the shear Σ_t (teal arrows) maps the graph of the force $F_t = -V_t'$ (navy) to a graph over the forecast ψ (gold), which is the graph of $-\Gamma_t'$. Right: the bare potential V_T, the effective potential V_t and Γ_t, each shifted to have minimum 0.

Theorem 8.4. (Wetterich equation) (formal) *Γ_t solves*

$$\partial_t \Gamma_t = \tfrac{1}{2}\mathrm{Tr}[\dot{R}_t\,((S_t^{-1} + D^2\Gamma_t)^{-1} - S_t)], \qquad \Gamma_T = V_T. \tag{8.12}$$

Equivalently, the effective average action $\Gamma_t^{\mathrm{EAA}}(\psi) := \Gamma_t(\psi) + \frac{1}{2}\langle \psi, Q_T^{-1}\psi\rangle$ *solves, up to a field-independent term,*

$$\partial_t \Gamma_t^{\mathrm{EAA}} = \tfrac{1}{2}\mathrm{Tr}[\dot{R}_t\,(D^2\Gamma_t^{\mathrm{EAA}} + R_t)^{-1}]. \tag{8.13}$$

It interpolates between the bare action $\Gamma_T^{\mathrm{EAA}} = \frac{1}{2}\langle \psi, Q_T^{-1}\psi\rangle + V_T(\psi)$ ($R_T = \infty$) and the 1PI effective action Γ_0^{EAA} of ν_T ($R_0 = 0$).

Proof. Integrate (8.8) in field space. With $F = D\Gamma$, the derivative of $H = (S^{-1} + D^2\Gamma)^{-1}$ in a direction v is $-H\,D^3\Gamma[v]\,H$, so by the total symmetry of $D^3\Gamma$,

$$D_v\mathrm{Tr}(\dot{R}_t H_t) = -\mathrm{Tr}(H_t \dot{R}_t H_t\, D^3\Gamma_t[v]) = -\mathrm{Tr}(C_t\, D^2 F_t)[v].$$

Hence (8.8) reads $D\left(\partial_t\Gamma_t - \frac{1}{2}\mathrm{Tr}(\dot{R}_t H_t)\right) = 0$, and (8.12) holds up to a function of t alone. That this function is $-\frac{1}{2}\mathrm{Tr}(\dot{R}_t S_t)$ with the normalisation (8.11) follows from the Polchinski equation by the envelope theorem (Exercise 8.5). For (8.13), note that $D^2\Gamma_t^{\mathrm{EAA}} + R_t = D^2\Gamma_t + S_t^{-1}$. For the endpoints: $S_T = 0$; and at $t = 0$, $S_0 = Q_T$, so (8.11) identifies $\frac{1}{2}\langle x, Q_T^{-1}x\rangle - V_0(x)$, up to a constant, with the cumulant generating function of ν_T at $Q_T^{-1}x$, whose Legendre transform is Γ_0^{EAA} (Exercise 8.7). □

Thus the Wetterich equation is the gradient-integrated form of the force equation (8.8), i.e. of the backward half of the FBSDE in forecast coordinates [Mor94, BCG26]. Every object in it is an observable of the Polchinski process:

Wetterich	FBSDE
field argument ψ	forecast $\psi_t = \mathbb{E}_t[\Phi_T]$, the state transported by Σ_t
$D\Gamma_t(\psi_t)$	minus the costate, $-F_t = \mathbb{E}_t[D\,V_T(\Phi_T)]$
propagator $(D^2\Gamma_t^{\mathrm{EAA}} + R_t)^{-1}$	conditional covariance Cov_t
regulator $R_t = S_t^{-1} - Q_T^{-1}$	precision of the unresolved shell, minus that of the prior
$\partial_t\Gamma_t$	$\frac{1}{2}\mathrm{Tr}[\dot{R}_t\,(\mathrm{Cov}_t - S_t)]$

The flow of Γ is driven by the difference between the interacting and the free conditional covariances, paired with the shell $\dot{R}_t$. It is *one-loop exact*, with the full propagator inside the trace. In the Gaussian case $D^2\Gamma_t$ is constant and only the constant term of Γ flows. For the heat profile with $T=\infty$, $S_t = C\,(1-\mathrm{e}^{-A/t^2})$ and

$$R_t = \frac{A}{\mathrm{e}^{A/t^2}-1}, \tag{8.14}$$

which behaves like a mass term t^2 on the modes $|k| \ll t$ (for $m \ll t$) and is exponentially small on the modes $|k| \gg t$: the exponential regulator of the functional renormalisation group (Figure 8.3).

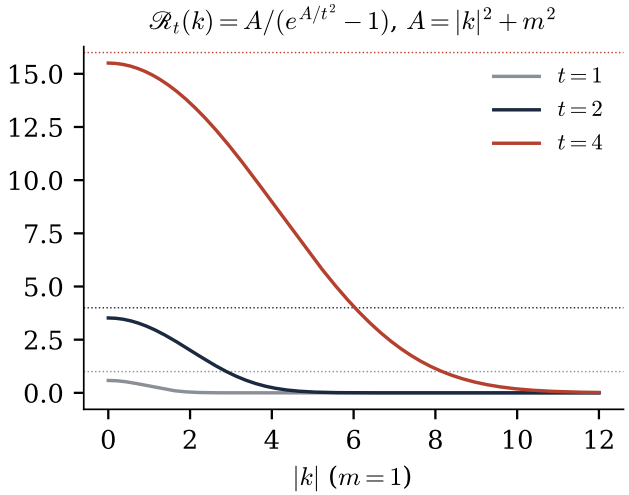


Figure 8.3. The regulator R_t of the heat profile ($T=\infty$, $m=1$): close to the mass term t^2 (dotted) on the modes $|k| \ll t$, exponentially small on the modes $|k| \gg t$.

8.6 Variational meaning and the tree level

Constrained free energies. (formal) By the Gibbs variational principle applied to the conditional law (7.5), and the identity $\mathrm{Ent}(\nu|\mu^S(\cdot - x)) = \mathrm{Ent}(\nu|\mu^S(\cdot-\psi)) + \frac{1}{2}\langle x-\psi, S^{-1}(x-\psi)\rangle$ for ν of mean ψ,

$$\Gamma_t(\psi) = \inf\Big\{\int V_T\,\mathrm{d}\nu + \mathrm{Ent}(\nu\,|\,\mu^{S_t}(\cdot-\psi)) \colon \textstyle\int \varphi\ \nu(\mathrm{d}\varphi) = \psi\Big\}. \tag{8.15}$$

So Γ_t, the effective average action up to the quadratic term $\frac{1}{2}\langle\psi, Q_T^{-1}\psi\rangle$, is the free energy of the unresolved shell constrained to the forecast ψ, and (8.11) is the minimisation over the forecast. In the language of Theorem 9.1: V_t is the value function of the control problem, and Γ_t is its Moreau transform, i.e. the value function expressed in the forecast coordinate, the interaction picture of the classical part of the dynamics.

The tree level. Dropping the Itô term in (8.8) gives $\Gamma_t = V_T$, and then (8.11) becomes the Hopf–Lax formula

$$V_t^{\mathrm{tree}}(x) = \inf_y \Big\{V_T(y) + \tfrac{1}{2}\langle y-x, S_t^{-1}(y-x)\rangle\Big\}, \tag{8.16}$$

the solution of the noiseless Polchinski equation $\partial_t V = \frac{1}{2}\langle DV, \dot{Q}\,DV\rangle$. It is the saddle point of the Gaussian convolution (7.3). Its minimiser solves the classical field equation $S_t^{-1}(y-x) + DV_T(y) = 0$ in the shell $[t,T]$ with background x, and expanding it in V_T produces exactly the tree diagrams.

Exercises

Exercise 8.1. (Stochastic maximum principle) (a) For the control problem (9.1), perturb an optimal control $u \to u + \varepsilon v$ and show that $p_t := \mathbb{E}_t[-DV_T(X_T^u)]$ satisfies $u_t = \dot{Q}_t^{1/2} p_t$. Identify p with the force F of (7.7). (b) Show that $Y_t = DF_t(\Phi_t)\,\dot{Q}_t^{1/2}$, and derive (8.3).

Exercise 8.2. (Characteristics) (a) Show that the transport equation $\partial_t F + DF[\dot{Q}F] = 0$ is solved by $F_t(x) = F_T(y)$, $y = x + S_t F_T(y)$, and that its characteristics are the trajectories of (8.1). (b) Verify that (8.16) solves $\partial_t V = \frac{1}{2}\langle DV, \dot{Q}\,DV\rangle$, and that it is the limit $\hbar \to 0$ of $-\hbar \log \int \mathrm{e}^{-V_T(x+\zeta)/\hbar}\mu^{\hbar S_t}(\mathrm{d}\zeta)$. (c) For $V_T(y) = \lambda y^4$ in zero dimensions, solve the equation for y as a power series in λ, and identify the coefficients as tree graphs.

Exercise 8.3. (Conditional moments) (a) Prove $\mathrm{Cov}(\mu_t^\phi) = S_t - S_t D^2 V_t(\phi)\, S_t$. (b) Prove (8.9).

Exercise 8.4. (A symplectic shear) (a) Show that Σ_t preserves $\sum_i \mathrm{d}x_i \wedge \mathrm{d}p_i$ iff S_t is symmetric. (b) Show that the image of the graph of $-DV_t$ under Σ_t is the graph of $-D\Gamma_t$, with Γ_t given by (8.11). (c) For a non-gradient force, show that Theorem 8.2 still holds, but that F_t need not be a gradient.

Exercise 8.5. (The constant) Differentiate (8.11) in t at fixed x (envelope theorem), and use (7.6) to show that the quadratic terms cancel and $\partial_t \Gamma_t = -\frac{1}{2}\mathrm{Tr}(\dot{Q}_t D^2 V_t)$. Deduce (8.12) with the stated constant.

Exercise 8.6. (Gaussian check) For $V_T(\varphi) = \frac{\kappa}{2}\langle\varphi,\varphi\rangle$, show that $\Gamma_t(\psi) = \frac{\kappa}{2}\langle\psi,\psi\rangle + \text{const}$, and solve the FBSDE (8.7) explicitly. Compare with Exercise 7.1.

Exercise 8.7. (1PI effective action) (a) Let $K(j) = \log \int \mathrm{e}^{\langle j,\varphi\rangle}\mathrm{e}^{-V_T(\varphi)}\mu^{Q_T}(\mathrm{d}\varphi)$. Show that $\Gamma_0^{\mathrm{EAA}} = K^*$ up to a constant, and that $D^2\Gamma_0^{\mathrm{EAA}}(\psi)$ is the inverse covariance of the tilted measure with mean ψ. (b) Integrate (8.12) with $D^2\Gamma_t$ replaced by $D^2 V_T$, and obtain the one-loop formula $\Gamma_t \approx V_T + \frac{1}{2}\mathrm{Trlog}\,(1 + S_t D^2 V_T)$.

Exercise 8.8. (Trees versus loops) Using (8.11), show that $V_t = \Gamma_t - \frac{1}{2}\langle D\Gamma_t, S_t D\Gamma_t\rangle + O(\lambda^3)$ for the Φ^4 interaction (7.1). Deduce from Proposition 10.2 that, to order λ^2, Γ_t is obtained from V_t by removing exactly the term $k=1$, the one-particle-reducible tree; the bubble ($k=2$), the sunset ($k=3$) and the vacuum term ($k=4$) are 1PI and remain. Where does the sunset come from in (8.12)?

Exercise 8.9. (Regulators) Derive (8.14). Compute R_t for the sharp Fourier profile, and for $\hat{Q}_t(k) = (|k|^2 + m^2)^{-1} - (|k|^2 + m^2 + t^2)^{-1}$ (whose kernels are not smooth, but the formulas make sense). In the last case show that $R_t = t^2$ when $T = \infty$: a mass regulator.

9 Variational structure

9.1 Finite volume: the Boué–Dupuis formula

Let $\mathbb{H}_a$ be the set of progressively measurable drifts $u\colon [0,T] \to L^2(\mathcal{M})$ with $\mathbb{E}\int_0^T \|u_s\|^2 \mathrm{d}s < \infty$, and set

$$I_t(u) := \int_0^t \dot{Q}_s^{1/2} u_s\,\mathrm{d}s, \qquad X^u := X + I(u),$$

the field driven by the control u under the Gaussian reference $\mathbb{P}_0$.

Theorem 9.1. (Boué–Dupuis [BD98]; Polchinski as HJB) (formal)

$$-\log \mathbb{E}_0[\mathrm{e}^{-V_T(X_T)}] = V_0(0) = \inf_{u\in\mathbb{H}_a} \mathbb{E}_0\Big[V_T(X_T^u) + \frac{1}{2}\int_0^T \|u_s\|^2 \mathrm{d}s\Big]. \tag{9.1}$$

The infimum is attained at the feedback control $u_s^* = -\dot{Q}_s^{1/2}\,DV_s(X_s^{u^*})$, and $\mathrm{Law}(X_T^{u^*}) = \nu_T$. The effective potential is the value function, and (7.6) is the Hamilton–Jacobi–Bellman equation of this control problem.

Proof. (Verification.) For any u, by Itô's formula and (7.6),

$$\begin{aligned}\mathrm{d}V_t(X_t^u) &= \Big[\tfrac{1}{2}\|\dot{Q}_t^{1/2} DV_t\|^2 + \langle \dot{Q}_t^{1/2} DV_t, u_t\rangle\Big]\mathrm{d}t + \mathrm{d}M_t \\ &= \Big[\tfrac{1}{2}\|u_t + \dot{Q}_t^{1/2} DV_t\|^2 - \tfrac{1}{2}\|u_t\|^2\Big]\mathrm{d}t + \mathrm{d}M_t,\end{aligned}$$

with $DV_t = DV_t(X_t^u)$ and a martingale M. Integrating and taking expectations,

$$\mathbb{E}_0\Big[V_T(X_T^u) + \frac{1}{2}\int_0^T \|u_t\|^2\mathrm{d}t\Big] = V_0(0) + \frac{1}{2}\,\mathbb{E}_0\int_0^T \|u_t + \dot{Q}_t^{1/2} DV_t(X_t^u)\|^2\mathrm{d}t. \tag{9.2}$$

So the cost is $\geq V_0(0)$, with equality iff $u = u^*$. With this control X^{u^*} solves the same SDE as Φ in (7.7). □

Euler–Lagrange equation = FBSDE. Perturbing $u \to u + \varepsilon v$ in (9.1), the first-order condition is $\mathbb{E}\int_0^T \langle u_s + \dot{Q}_s^{1/2}\,\mathbb{E}_s[D V_T(X_T^u)], v_s\rangle \mathrm{d}s = 0$ for all adapted v, i.e.

$$u_s = -\dot{Q}_s^{1/2}\,\mathbb{E}_s[D V_T(X_T^u)] = \dot{Q}_s^{1/2} F_s. \tag{9.3}$$

This is (7.8). The FBSDE is the Euler–Lagrange equation of the variational problem, and the Polchinski equation is its Hamilton–Jacobi equation. These are the three faces of the same object: *the PDE* (Polchinski), *the stochastic dynamics* (FBSDE), and *the variational principle* (Boué–Dupuis).

Entropy. By Girsanov, $\frac{1}{2}\mathbb{E}_0\int \|u\|^2 \geq \mathrm{Ent}(\mathrm{Law}(X^u)\,|\,\mathrm{Law}(X))$, which is at least the entropy of the terminal marginal. So (9.1) is a pathwise lift of the Gibbs variational principle $-\log Z_T = \inf_\nu\{\int V_T\,\mathrm{d}\nu + \mathrm{Ent}(\nu|\mu^{Q_T})\}$. The lift is useful because adapted drifts can be *constructed scale by scale*: one looks for u as a functional of the Gaussian process X, not as an unknown measure.

Generating functionals. Adding a test functional f,

$$\mathcal{W}_T(f) := -\log\int \mathrm{e}^{-f}\mathrm{d}\nu_T = \inf_u \mathbb{E}_0\left[f(X_T^u) + V_T(X_T^u) + \frac{1}{2}\int_0^T \|u_s\|^2 \mathrm{d}s\right] - V_0(0). \tag{9.4}$$

Bounds on $\mathcal{W}_T(f)$ uniform in T, for f in a rich class, give tightness, moment bounds and the identification of limits. For Φ_3^4 the naive problem does not have uniform bounds: the leading-order optimal drift $u_s \approx -4\lambda\,\dot{Q}_s^{1/2}{:}X_s^3{:}$ has a divergent cost. One renormalises by the change of variables $u = -4\lambda\,\dot{Q}^{1/2}{:}X^3{:} + r$. After it, and after further paracontrolled changes of variables, the singular terms are explicit polynomials in the Wick powers of X, cancelled by *the counterterms*, and the residual functional is coercive. Its control in Besov and paracontrolled norms, uniformly in T, is the construction of Φ_3^4 in [BG20]; see also [BG21b].

9.2 Infinite volume

On $\mathbb{T}_L^d$ the free energy $V_0(0)$ and the effective potential V_t are extensive: they grow like L^d, and (9.1) has no limit as $L \to \infty$. Three objects are intensive and survive formally:

i. *the force.* For a local interaction $V_T(\varphi) = \int v_T(\varphi(x))\mathrm{d}x$, $D V_T(\varphi)(x) = v_T'(\varphi(x))$, and (7.8) reads, on $\mathbb{R}^d$,

$$\Phi_t = \int_0^t \dot{Q}_s F_s\,\mathrm{d}s + \int_0^t \dot{Q}_s^{1/2}\mathrm{d}W_s, \qquad F_t(x) = -\mathbb{E}_t[v_T'(\Phi_T(x))]. \tag{9.5}$$

Here every quantity is local, and the equation makes sense without a volume cutoff. Infinite-volume limits of $\nu_T^{(L)}$ are expected to be laws of solutions of (9.5). Uniqueness of such solutions (e.g. for small coupling, or by contraction in weighted spaces) is a uniqueness statement for the infinite-volume theory. Non-uniqueness is expected in the presence of phase transitions, e.g. for Φ_2^4 with a deep double-well potential [GJS75];

ii. *the Burgers flow* (7.12) of the force, for the same reason;

iii. *differences of free energies.* For a local f, $\mathcal{W}_T^{(L)}(f)$ in (9.4) is a difference of two extensive infima, and it stays bounded as $L \to \infty$. Its variational characterisation, through the infinite-volume Euler–Lagrange equation (9.5), is the basis of the infinite-volume analysis of Φ_2^4 and $\exp(\Phi)_2$ in [BG24].

The coercive estimates must then be *local*: one uses weighted norms and needs bounds whose constants do not depend on the volume. This infrared problem is the analogue, for the scale dynamics, of the "coming down from infinity" estimates for the parabolic dynamics.

Exercises

Exercise 9.1. **(Optimal drift)** (a) Derive (9.3) from (9.1), and check that the optimal u is a martingale up to the factor $\dot{Q}^{1/2}$. (b) For the quadratic interaction of Exercise 7.1, compute the optimal drift and the value (9.1) explicitly.

Exercise 9.2. **(Cost of the leading drift)** For $V_T = \lambda \int {:}\varphi^4{:}_T$ on $\mathbb{T}^3_L$, show that

$$\frac{1}{2}\,\mathbb{E}\int_0^T \|4\lambda\,\dot{Q}_s^{1/2}{:}X_s^3{:}\|^2 \mathrm{d}s = 12\lambda^2 L^3 \int Q_T(z)^4\,\mathrm{d}z,$$

which diverges linearly in T. Relate this to the $k=4$ term of (10.6).

Part II. Additional material

The following sections complement the main lecture. They develop the algebraic and perturbative structure of the Polchinski flow, and the Dyson–Schwinger equations at every scale.

10 Algebraic structure of the Polchinski flow

10.1 A semigroup of renormalisation maps

For a positive covariance S define the renormalisation map

$$\mathcal{R}_S(V) := -\log\,(\mu^S * \mathrm{e}^{-V}), \qquad (\mu^S * u)(\phi) = \int u\,(\phi+\zeta)\,\mu^S(\mathrm{d}\zeta). \tag{10.1}$$

Since $\mu^S * \mu^{S'} = \mu^{S+S'}$,

$$\mathcal{R}_S \circ \mathcal{R}_{S'} = \mathcal{R}_{S+S'} = \mathcal{R}_{S'} \circ \mathcal{R}_S, \qquad V_t = \mathcal{R}_{Q_{T,t}}(V_T). \tag{10.2}$$

So the effective potentials are an orbit of an *abelian* semigroup, indexed by the cone of positive covariances, acting nonlinearly on interactions and linearly on Boltzmann factors e^{-V}. Its infinitesimal generators are the vector fields

$$\mathcal{D}_B(V) := \tfrac{1}{2}\mathrm{Tr}(B\,D^2 V) - \tfrac{1}{2}\langle DV, B\,DV\rangle, \qquad \tfrac{\mathrm{d}}{\mathrm{d}s}\mathcal{R}_{sB}(V)\Big|_{s=0} = \mathcal{D}_B(V), \tag{10.3}$$

and (7.6) reads $\partial_t V_t = -\mathcal{D}_{\dot{Q}_t}(V_t)$.

Proposition 10.1. **(Flatness)** (formal) *(a)* $[\mathcal{D}_B, \mathcal{D}_{B'}] = 0$ *for all positive* B, B'. *(b)* V_t *depends on the profile only through* $Q_{T,t}$. *In particular the free energy* $V_0(0)$ *and the law* ν_T *depend only on* Q_T, *not on the path* $t \mapsto Q_t$.

Proof. (a) Differentiate $\mathcal{R}_{sB}\,\mathcal{R}_{rB'} = \mathcal{R}_{rB'}\,\mathcal{R}_{sB}$ at $s = r = 0$, or compute directly (Exercise 10.1). (b) is (10.2). □

Different profiles with the same endpoints are different *coordinate systems* on the same renormalisation group trajectory. They change the process Φ and the observables at intermediate resolution, but not the terminal theory. The flow has no holonomy [BCG26].

10.2 Perturbation theory and counterterms

Expanding $V_t = \mathcal{R}_{Q_{T,t}}(V_T)$ in cumulants of the Gaussian increment $\zeta \sim \mu^{Q_{T,t}}$ gives

$$V_t(\phi) = \mathbb{E}_\zeta\,[V_T(\phi+\zeta)] - \tfrac{1}{2}\mathrm{Var}_\zeta(V_T(\phi+\zeta)) + \dots \tag{10.4}$$

For (7.1), Exercise 6.1 of Lecture 3 shows that Gaussian convolution simply *lowers the Wick ordering*: since $Q_{T,t}(x,x) = c_T - c_t$,

$$V_t(\phi) = \int [\lambda H_4(\phi(x); c_t) + \gamma_T H_2(\phi(x); c_t)]\mathrm{d}x + O(\lambda^2). \tag{10.5}$$

Here $\gamma_T = O(\lambda^2)$ is kept for later use. At first order the effective potential is the Wick-ordered potential at the current resolution. This is the dual statement of Proposition 6.1 of Lecture 3: $\int H_4(X_t; c_t)$ is an observable.

Proposition 10.2. (Second order) (formal) *Write* $S = Q_{T,t}$ *and* $:\phi^p:_t = H_p(\phi; c_t)$. *Then*

$$V_t(\phi) = \int [\lambda :\phi^4:_t + \gamma_T :\phi^2:_t] - \frac{\lambda^2}{2} \sum_{k=1}^{4} \binom{4}{k}^2 k! \iint S(x,y)^k :\phi^{4-k}:_t (x) :\phi^{4-k}:_t (y)\, \mathrm{d}x\, \mathrm{d}y + O(\lambda^3). \tag{10.6}$$

Proof. By the binomial formula for Hermite polynomials,

$$H_4(\phi + \zeta; c_t + s) = \sum_{k=0}^{4} \binom{4}{k} H_k(\zeta; s)\, H_{4-k}(\phi; c_t), \qquad s = S(x,x).$$

Then compute the variance in (10.4) with Lemma 3.8 of Lecture 2. □

Now let $T \to \infty$ at fixed t, so that $S \to C - Q_t$, which near the diagonal is as singular as G. In $d = 2$ every S^k is locally integrable, so with $\gamma_T = 0$ every term has a limit: Wick ordering suffices. In $d = 3$, $G(z) \sim (4\pi|z|)^{-1}$:

- $k = 1, 2$: S and S^2 are locally integrable. These terms are finite and nonlocal.
- $k = 3$ (the *sunset*): $S^3 \sim |z|^{-3}$ is not integrable. Writing $\phi(x)\,\phi(y) = \frac{1}{2}(\phi(x)^2 + \phi(y)^2) - \frac{1}{2}(\phi(x) - \phi(y))^2$, up to a field-independent constant the divergent part is local: $-48\lambda^2(\int S^3) \int :\phi^2:_t$ + (finite), with $\int S^3 \sim \frac{1}{16\pi^2} \log T$. It is cancelled by

  $$\gamma_T := 48\lambda^2 \int_{\mathbb{T}_L^3} Q_T(z)^3\, \mathrm{d}z = \frac{3\lambda^2}{\pi^2} \log T + O(1), \tag{10.7}$$

  since $\gamma_T - 48\lambda^2 \int S^3$ converges (Exercise 10.2).
- $k = 4$: a divergent constant $-12\lambda^2 L^3 \int S^4 \sim T$, the vacuum energy. It cancels in ν_T but not in the free energy.

Power counting (Exercise 10.3) shows that no further field-dependent counterterms are needed in $d = 3$: Φ_3^4 is superrenormalisable. Thus renormalisation appears as a property of the flow: one chooses the terminal data V_T so that the effective potentials V_t at *fixed* resolution have limits. The divergences at second order are also related to the singularity of Φ_3^4 with respect to the GFF [BG21b]: the measures ν_t at fixed t are expected to stay absolutely continuous with respect to μ^{Q_t}, while the terminal one is not.

Exercises

Exercise 10.1. (Commuting flows) Show by direct computation that $D\,\mathcal{D}_B(V)[\mathcal{D}_{B'}(V)] = D\,\mathcal{D}_{B'}(V)[\mathcal{D}_B(V)]$. Which property of D^2V and D^3V is used?

Exercise 10.2. (The sunset) (a) Show that $\int_{|z|\le 1} G(z)^3\,\mathrm{d}z = \infty$ in $d = 3$, and that $\int Q_T^3 = \frac{1}{16\pi^2}\log T + O(1)$ for the heat profile. (b) Show that $Q_T^3 - (Q_T - Q_t)^3$ is integrable uniformly in T, and deduce that the counterterm (10.7) makes the $:\phi^2:$ coefficient in (10.6) converge. (c) Show that the remaining part $\iint S^3(x,y)\,(\phi(x) - \phi(y))^2$ converges for smooth ϕ.

Exercise 10.3. (**Power counting**) Show that the superficial degree of divergence of a connected graph of Φ^4_d with n vertices and E external legs is $\omega = d - \frac{d-2}{2}E + (d-4)\,n$. Deduce that in $d=3$ the only divergent graphs with loops are the two-point graphs with $n \le 2$ (the tadpole and double tadpole, removed by Wick ordering, and the sunset) and vacuum graphs with $n \le 3$.

11 Dyson–Schwinger equations at every scale

Assume now that all the operators involved are functions of A, hence commute, and let

$$K := Q_T^{-1}, \qquad J_t := K Q_t, \qquad J_0 = 0, \quad J_T = 1.$$

For the heat profile, $K = A e^{A/T^2}$ and $J_t = e^{-A(t^{-2}-T^{-2})}$ is a heat semigroup, which tends to e^{-A/t^2} as $T \to \infty$. F_t is the force observable of Theorem 7.4.

Theorem 11.1. (**DS equations at resolution t**) (formal) *The* noise observable

$$\eta_t := K\,\Phi_t - J_t F_t \tag{11.1}$$

is a centred $\mathbb{P}$-martingale, with $\mathrm{d}\eta_t = K \dot{Q}_t^{1/2} \mathrm{d}W_t^{\mathbb{P}} - J_t\,\mathrm{d}F_t$. For every $t \le T$, every h and every functional G,

$$\mathbb{E}^{\mathbb{P}}\left[D_{J_t h} G(\Phi_t)\right] = \mathbb{E}^{\mathbb{P}}\left[G(\Phi_t)\,\langle h, \eta_t\rangle\right] = \mathbb{E}^{\mathbb{P}}\left[G(\Phi_t)\,\langle h, K\,\Phi_t - J_t F_t\rangle\right]. \tag{11.2}$$

At $t = T$ this is the Dyson–Schwinger equation of ν_T,

$$\mathbb{E}^{\nu_T}\left[D_h G\right] = \mathbb{E}^{\nu_T}\left[G\,\langle h, K\,\varphi + D\,V_T(\varphi)\rangle\right]. \tag{11.3}$$

Proof. From (7.7) and $\dot{J}_t = K\,\dot{Q}_t$, $\mathrm{d}\eta_t = K\,(\dot{Q}_t F_t \mathrm{d}t + \dot{Q}_t^{1/2}\mathrm{d}W_t^{\mathbb{P}}) - K\,\dot{Q}_t F_t \mathrm{d}t - J_t \mathrm{d}F_t$. The drift cancels, F is a martingale and $\eta_0 = 0$. For (11.2) use Proposition 7.2: $\Phi_t \sim \nu_t \propto e^{-V_t}\mu^{Q_t}$. Gaussian integration by parts for μ^{Q_t} in the direction $Q_t k$ gives $\mathbb{E}^{\nu_t}[D_{Q_t k} G] = \mathbb{E}^{\nu_t}[G\,(\langle k, \varphi\rangle + \langle Q_t k, D\,V_t(\varphi)\rangle)]$. Take $k = Kh$, so that $Q_t k = J_t h$, and recall $F_t = -D\,V_t(\Phi_t)$. □

Formally letting $T \to \infty$, $K \to A$, and (11.1) becomes an exact identity between observables at every resolution:

$$A\,\Phi_t = A\,Q_t\,F_t + \eta_t, \tag{11.4}$$

with $A\,Q_t = e^{-A/t^2}$ for the heat profile. This is the Euclidean field equation $A\varphi = -V'(\varphi) + \text{noise}$ of Lectures 1–2, seen at resolution t. It needs no products of distributions: the nonlinearity enters only through the force *observable* F_t, smeared at scale $1/t$. For the GFF ($F = 0$), (11.2) is Gaussian integration by parts for μ^{Q_t}. For Φ^4_3, where F_t plays the role of the renormalised cube, the limit $t \to \infty$ of (11.4) corresponds to the renormalised DS equation of [GH21].

Remark 11.2. (**Comparison with stochastic quantisation**) In parabolic or elliptic stochastic quantisation (Lecture 3, §4) the field equation is imposed in an auxiliary time, and the field at fixed time has the right law. Here the field equation (11.4) holds at each resolution, and the auxiliary parameter is the resolution itself. Taking conditional expectations of an SPDE solution with respect to a scale filtration gives identities of the same shape, but with a different noise covariance: the martingale $J_t\,\mathrm{d}F_t$ in $\mathrm{d}\eta_t$ is essential [BCG26].

Exercises

Exercise 11.1. (**DS from the dynamics**) Prove (11.2) without using Proposition 7.2, for G of the form $G_t(\Phi_t)$ with G_t solving (7.10). Show that the drift of $D_{J_t h} G_t(\Phi_t)$ equals the covariation of $G_t(\Phi_t)$ with $\langle h, \eta_t\rangle$. Where is the symmetry of $D F_t$ used?

Exercise 11.2. (**Gaussian check**) For $V_T = 0$ show that (11.2) reduces to $\mathbb{E}[D_{Q,Kh} G(X_t)] = \mathbb{E}[G(X_t)\langle Kh, X_t\rangle]$, and compare with the DS characterisation of the GFF in Lecture 2.

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